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Mathematical Background

Preview

The principal aim of large-sample theory is to provide simple approximations for quantities that are difficult to calculate exactly. The approach throughout the book is to embed the actual situation in a sequence of situations, the limit of which serves as the desired approximation.

The present chapter reviews some of the basic ideas from calculus required for this purpose such as limit, convergence of a series, and continuity. Section 1 defines the limit of a sequence of numbers and develops some of the properties of such limits. In Section 2, the embedding idea is introduced and is illustrated with two approximations of binomial probabilities. Section 3 provides a brief introduction to infinite series, particularly power series. Section 4 is concerned with different rates at which sequences can tend to infinity (or zero); it introduces the o , $\asymp$, and O notation and the three most important growth rates: exponential, polynomial, and logarithmic. Section 5 extends the limit concept to continuous variables, defines continuity of a function, and discusses the fact that monotone functions can have only simple discontinuities. This result is applied in Section 6 to cumulative distribution functions; the section also considers alternative representations of probability distributions and lists the densities of probability functions of some of the more common distributions.

1.1 The concept of limit

Large-sample (or asymptotic*) theory deals with approximations to probability distributions and functions of distributions such as moments and quantiles. These approximations tend to be much simpler than the exact formulas and, as a result, provide a basis for insight and understanding that often would be difficult to obtain otherwise. In addition, they make possible simple calculations of critical values, power of tests, variances of estimators, required sample sizes, relative efficiencies of different methods, and so forth which, although approximate, are often accurate enough for the needs of statistical practice.

Underlying most large-sample approximations are limit theorems in which the sample sizes tend to infinity. In preparation, we begin with a discussion of limits. Consider a sequence of numbers a_n such as

$$(1.1.1) \quad a_n = 1 - \frac{1}{n} (n = 1, 2, \dots): \quad 0, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}, \dots,$$

and

$$(1.1.2) \quad a_n = 1 - \frac{1}{n^2} (n = 1, 2, \dots): \quad 0, \frac{3}{4}, \frac{8}{9}, \frac{15}{16}, \frac{24}{25}, \frac{35}{36}, \dots,$$

or, more generally, the sequences

$$(1.1.3) \quad a_n = a - \frac{1}{n} \text{ and } a_n = a - \frac{1}{n^2}$$

for some arbitrary fixed number a .

Two facts seem intuitively clear: (i) the members of both sequences in (1.1.3) are getting arbitrarily close to a as n gets large; (ii) this “convergence” toward a proceeds faster for the second series than for the first. The present chapter will make these two concepts precise and give some simple applications. But first, consider some additional examples.

The sequence obtained by alternating members of the two sequences (1.1.3) is given by

$$(1.1.4) \quad a_n = \begin{cases} a - \frac{1}{n} & \text{if } n \text{ is odd,} \\ a - \frac{1}{n^2} & \text{if } n \text{ is even:} \end{cases}$$

$$a - 1, a - \frac{1}{4}, a - \frac{1}{3}, a - \frac{1}{16}, a - \frac{1}{5}, a - \frac{1}{36}, \dots$$

*The term “asymptotic” is not restricted to large-sample situations but is used quite generally in connection with any limit process. See, for example, Definition 1.1.3. For some general discussion of asymptotics, see, for example, DeBruijn (1958).

For this sequence also, the numbers get arbitrarily close to a as n gets large. However, they do so without each member being closer to a than the preceding one. For a sequence $a_n, n = 1, 2, \dots$, to tend to a limit a as $n \rightarrow \infty$, it is not necessary for each a_n to be closer to a than its predecessor a_{n-1} , but only for a_n to get arbitrarily close to a as n gets arbitrarily large.

Let us now formalize the statement that the members of a sequence $a_n, n = 1, 2, \dots$, get arbitrarily close to a as n gets large. This means that for any interval about a , no matter how small, the members of the sequence will eventually, i.e., from some point on, lie in the interval. If such an interval is denoted by $(a - \epsilon, a + \epsilon)$ the statement says that from some point on, i.e., for all n exceeding some n_0 , the numbers a_n will satisfy $a - \epsilon < a_n < a + \epsilon$ or equivalently

$$(1.1.5) \quad |a_n - a| < \epsilon \text{ for all } n > n_0.$$

The value of n_0 will of course depend on ϵ , so that we will sometimes write it as $n_0(\epsilon)$; the smaller ϵ is, the larger is the required value of $n_0(\epsilon)$.

Definition 1.1.1 The sequence $a_n, n = 1, 2, \dots$, is said to tend (or converge) to a limit a ; in symbols:

$$(1.1.6) \quad a_n \rightarrow a \text{ as } n \rightarrow \infty \quad \text{or} \quad \lim_{n \rightarrow \infty} a_n = a$$

if, given any $\epsilon > 0$, no matter how small, there exists $n_0 = n_0(\epsilon)$ such that (1.1.5) holds.

For a formal proof of a limit statement (1.1.6) for a particular sequence a_n , it is only necessary to produce a value $n_0 = n_0(\epsilon)$ for which (1.1.5) holds. As an example consider the sequence (1.1.1). Here $a = 1$ and $a_n - a = -1/n$. For any given ϵ , (1.1.5) will therefore hold as soon as $\frac{1}{n} < \epsilon$ or $n > \frac{1}{\epsilon}$. For $\epsilon = 1/10, n_0 = 10$ will do; for $\epsilon = 1/100, n_0 = 100$; and, in general, for any ϵ , we can take for n_0 the smallest integer, which is $\geq \frac{1}{\epsilon}$.

In examples (1.1.1)–(1.1.4), the numbers a_n approach their limit from one side (in fact, in all these examples, $a_n < a$ for all n). This need not be the case, as is shown by the sequence

$$(1.1.7) \quad a_n = \begin{cases} 1 - \frac{1}{n} & \text{if } n \text{ is odd} \\ 1 + \frac{1}{n} & \text{if } n \text{ is even} \end{cases} = 1 + (-1)^n \frac{1}{n}.$$

It may be helpful to give an example of a sequence which does not tend to a limit. Consider the sequence

$$0, 1, 0, 1, 0, 1, \dots$$

given by $a_n = 0$ or 1 as n is odd or even. Since for arbitrarily large n, a_n takes on the values 0 and 1 , it cannot get arbitrarily close to any a for all sufficiently large n .

The following is an important example which we state without proof.

Example 1.1.1 The exponential limit. For any finite number c ,

$$(1.1.8) \quad \left(1 + \frac{c}{n}\right)^n \rightarrow e^c \text{ as } n \rightarrow \infty.$$

To give an idea of the speed of the convergence of $a_n = \left(1 + \frac{1}{n}\right)^n$ to its limit e , here are the values of a_n for a number of values of n , and the limiting value $e(n = \infty)$ to the nearest 1/100.

TABLE 1.1.1. $\left(1 + \frac{1}{n}\right)^n$ to the nearest 1/100

n	1	3	5	10	30	50	100	500	∞
a_n	2.00	2.37	2.49	2.59	2.67	2.69	2.70	2.72	2.72

To the closest 1/1000, one has $a_{500} = 2.716$ and $e = 2.718$. □

The idea of limit underlies all of large-sample theory. Its usefulness stems from the fact that complicated sequences $\{a_n\}$ often have fairly simple limits which can then be used to approximate the actual a_n at hand. Table 1.1.1 provides an illustration (although here the sequence is fairly simple). It suggests that the limit value $a = 2.72$ shown in Table 1.1.1 provides a good approximation for $n \geq 30$ and gives a reasonable ballpark figure even for n as small as 5.

Contemplation of the table may raise a concern. There is no guarantee that the progress of the sequence toward its limit is as steady as the tabulated values suggest. The limit statement guarantees only that *eventually* the members of the sequence will be arbitrarily close to the limit value, not that each member will be closer than its predecessor. This is illustrated by the sequence (1.1.4). As another example, let

$$(1.1.9) \quad a_n = \begin{cases} 1/\sqrt{n} & \text{if } n \text{ is the square of an integer } (n = 1, 4, 9, \dots) \\ 1/n & \text{otherwise.} \end{cases}$$

Then $a_n \rightarrow 0$ (Problem 1.7) but does so in a somewhat irregular fashion. For example, for $n = 90, 91, \dots, 99$, we see a_n getting steadily closer to the limit value 0 only to again be substantially further away at $n = 100$. In sequences encountered in practice, such irregular behavior is rare. (For a statistical example in which it does occur, see Hodges (1957)). A table such as Table 1.1.1 provides a fairly reliable indication of smooth convergence to the limit.

Limits satisfy simple relationships such as: if $a_n \rightarrow a, b_n \rightarrow b$, then

$$(1.1.10) \quad a_n + b_n \rightarrow a + b \quad \text{and} \quad a_n - b_n \rightarrow a - b,$$

$$(1.1.11) \quad a_n \cdot b_n \rightarrow a \cdot b$$

and

$$(1.1.12) \quad a_n/b_n \rightarrow a/b \text{ provided } b \neq 0.$$

These results will not be proved here. Proofs and more detailed treatment of the material in this section and Section 1.3 are given, for example, in the classical texts (recently reissued) by Hardy (1992) and Courant (1988). For a slightly more abstract treatment, see Rudin (1976).

Using (1.1.12), it follows from (1.1.8), for example, that

$$(1.1.13) \quad \left(\frac{1 + \frac{a}{n}}{1 + \frac{b}{n}} \right)^n \rightarrow e^{a-b} \text{ as } n \rightarrow \infty.$$

An important special case not covered by Definition 1.1.1 arises when a sequence tends to ∞ . We say that $a_n \rightarrow \infty$ if eventually (i.e., from some point on) the a 's get larger than any given constant M . Proceeding as in Definition 1.1.1, this leads to

Definition 1.1.2 The sequence a_n tends to ∞ ; in symbols,

$$(1.1.14) \quad a_n \rightarrow \infty \quad \text{or} \quad \lim_{n \rightarrow \infty} a_n = \infty$$

if, given any M , no matter how large, there exists $n_0 = n_0(M)$ such that

$$(1.1.15) \quad a_n > M \text{ for all } n > n_0.$$

Some sequences tending to infinity are

$$(1.1.16) \quad a_n = n^\alpha \text{ for any } \alpha > 0$$

(this covers sequences such as $\sqrt[3]{n} = n^{1/3}$, $\sqrt{n} = n^{1/2}$, $\dots$ and $n^2, n^3, \dots$);

$$(1.1.17) \quad a_n = e^{\alpha n} \text{ for any } \alpha > 0;$$

$$(1.1.18) \quad a_n = \log n, \quad a_n = \sqrt{\log n}, \quad a_n = \log \log n.$$

To see, for example, that $\log n \rightarrow \infty$, we check (1.1.15) to find that $\log n > M$ provided $n > e^M$ (here we use the fact that $e^{\log n} = n$), so that we can take for n_0 the smallest integer that is $\geq e^M$.

Relations (1.1.10)–(1.1.12) remain valid even if a and/or b are $\pm\infty$ with the exceptions that $\infty - \infty$, $\infty \cdot 0$, and ∞/∞ are undefined.

The case $a_n \rightarrow -\infty$ is completely analogous (Problem 1.4) and requires the corresponding restrictions on (1.1.10)–(1.1.12).

Since throughout the book we shall be dealing with sequences, we shall in the remainder of the present section and in Section 4 consider relations between two sequences a_n and b_n , $n = 1, 2, \dots$, which are rough analogs of the relations $a = b$ and $a < b$ between numbers.

Definition 1.1.3 Two sequences $\{a_n\}$ and $\{b_n\}$ are said to be (asymptotically) equivalent as $n \rightarrow \infty$; in symbols:

$$(1.1.19) \quad a_n \sim b_n$$

if

$$(1.1.20) \quad a_n/b_n \rightarrow 1.$$

This generalizes the concept of equality of two numbers a and b , to which it reduces for the sequences $a, a, \dots$ and $b, b, \dots$.

If b_n tends to a finite limit $b \neq 0$, (1.1.20) simply states that a_n tends to the same limit. However, if the limit b is 0 or $\pm\infty$, the statement $a_n \sim b_n$ contains important additional information. Consider, for example, the sequences $a_n = 1/n^2$ and $b_n = 1/n$, both of which tend to zero. Since their ratio a_n/b_n tends to zero, the two sequences are not equivalent. Here are two more examples:

$$(1.1.21) \quad a_n = n + n^2, \quad b_n = n$$

and

$$(1.1.22) \quad a_n = n + n^2, \quad b_n = n^2,$$

in both of which a_n and b_n tend to ∞ . In the first, $a_n/b_n \rightarrow \infty$ so that a_n and b_n are not equivalent; in the second, $a_n/b_n \rightarrow 1$ so that they are equivalent.

A useful application of the idea of equivalence is illustrated by the sequences

$$(1.1.23) \quad a_n = \frac{1}{n} + \frac{3}{n^2} + \frac{1}{n^3}, \quad b_n = \frac{1}{n}.$$

Both a_n and b_n tend to zero. Since their ratio satisfies

$$\frac{a_n}{b_n} = 1 + \frac{3}{n} + \frac{1}{n^2} \rightarrow 1,$$

the two sequences are equivalent. The replacement of a complicated sequence such as a_n by a simpler asymptotically equivalent sequence b_n plays a central role in large-sample theory.

Replacing a true a_n by an approximating b_n of course results in an error. Consider, for example, the two equivalent sequences (1.1.22). When $n = 100$,

$$a_n = 10,100, \quad b_n = 10,000,$$

and the error (or absolute error) is $|a_n - b_n| = 100$. On the other hand, the

$$(1.1.24) \quad \text{relative error} = \left| \frac{a_n - b_n}{a_n} \right|$$

is $\frac{100}{10,100}$ which is less than .01. The small relative error corresponds to the fact that, despite the large absolute error of 100, b_n gives a pretty good idea of the size of a_n .

As the following result shows, asymptotic equivalence is closely related to relative error.

Lemma 1.1.1 *The sequences $\{a_n\}$ and $\{b_n\}$ are asymptotically equivalent if and only if the relative error tends to zero.*

Proof. The relative error

$$\left| \frac{a_n - b_n}{a_n} \right| = \left| 1 - \frac{b_n}{a_n} \right| \rightarrow 0$$

if and only if $b_n/a_n \rightarrow 1$. ■

The following is a classical example of asymptotic equivalence which forms the basis of the application given in the next section.

Example 1.1.2 Stirling's formula. Consider the sequence

$$(1.1.25) \quad a_n = n! = 1 \cdot 2 \cdots n.$$

Clearly, $a_n \rightarrow \infty$ as $n \rightarrow \infty$, but it is difficult from the defining formula to see how fast this sequence grows. We shall therefore try to replace it by a simpler equivalent sequence b_n . Since n^n is clearly too large, one might try, for example, $(n/2)^n$. This turns out to be still too large, but taking logarithms leads (not obviously) to the suggestion $b_n = (n/e)^n$. Now only a relatively minor further adjustment is required, and the final result (which we shall not prove) is Stirling's formula

$$(1.1.26) \quad n! \sim \sqrt{2\pi n} (n/e)^n.$$

The following table adapted from Feller (Vol. 1) (1957), where there is also a proof of (1.1.26), shows the great accuracy of the approximation (1.1.26) even for small n .

It follows from Lemma 1.1.1 that the relative error tends to zero, and this is supported by the last line of the table. On the other hand, the absolute error tends to infinity and is already about 30,000 for $n = 10$. □

The following example provides another result, which will be used later.

TABLE 1.1.2. Stirling's approximation to $n!$

n	1	2	5	10	100
$n!$	1	2	120	3.6288×10^6	9.3326×10^{157}
(1.26)	.922	1.919	118.019	3.5987×10^6	9.3249×10^{157}
Error	.078	.081	1.981	$.0301 \times 10^6$	$.0077 \times 10^{157}$
Relative Error	.08	.04	.02	.008	.0008

Example 1.1.3 Sums of powers of integers. Let

$$(1.1.27) \quad S_n^{(k)} = 1^k + 2^k + \cdots + n^k \quad (k \text{ a positive integer})$$

so that, in particular,

$$S_n^{(0)} = n, \quad S_n^{(1)} = \frac{n(n+1)}{2}, \quad \text{and} \quad S_n^{(2)} = \frac{n(n+1)(2n+1)}{6}.$$

These formula suggest that perhaps

$$(1.1.28) \quad S_n^{(k)} \sim \frac{n^{k+1}}{k+1} \quad \text{for all } k = 1, 2, \dots,$$

and this is in fact the case. (For a proof, see Problem 1.14). $\square$

Summary

1. A sequence of numbers a_n tends to the *limit* a if for all sufficiently large n the a 's get arbitrarily close [i.e., within any preassigned distance ε] to a . If $a_n \rightarrow a$, then a can be used as an approximation for a_n when n is large.
2. Two sequences $\{a_n\}$ and $\{b_n\}$ are *asymptotically equivalent* if their ratio tends to 1. The members of a complicated sequence can often be approximated by those of a simpler sequence which is asymptotically equivalent. In such an approximation, the relative error tends to 0 as $n \rightarrow \infty$.
3. Stirling's formula provides a simple approximation for $n!$. The relative error in this approximation tends to 0 as $n \rightarrow \infty$ while the absolute error tends to ∞ .

1.2 Embedding sequences

The principal aim of the present section is to introduce a concept which is central to large-sample theory: obtaining an approximation to a given

situation by embedding it in a suitable sequence of situations. We shall illustrate this process by obtaining two different approximations for binomial probabilities corresponding to two different embeddings.

The probability of obtaining x successes in n binomial trials with success probability p is

$$(1.2.1) \quad P_n(x) = \binom{n}{x} p^x q^{n-x} \quad \text{where } q = 1 - p.$$

Suppose that n is even and that we are interested in the probability $P_n\left(\frac{n}{2}\right)$ of getting an even split between successes and failures. It seems reasonable to expect that this probability will tend to 0 as $n \rightarrow \infty$ and that it will be larger when $p = 1/2$ than when it is $\neq 1/2$.

To get a more precise idea of this behavior, let us apply Stirling's formula (1.1.26) to the three factorials in

$$P_n\left(\frac{n}{2}\right) = \frac{n!}{\left(\frac{n}{2}\right)! \left(\frac{n}{2}\right)!} (pq)^{n/2}.$$

After some simplification, this leads to (Problem 2.1)

$$(1.2.2) \quad P_n\left(\frac{n}{2}\right) \sim \sqrt{\frac{2}{\pi}} \cdot \frac{1}{\sqrt{n}} \left(\sqrt{4pq}\right)^n.$$

We must now distinguish two cases.

Case 1. $p = 1/2$. Here we are asking for an even split between heads and tails in n tosses with a fair coin. The third factor in (1.2.2) is then 1, and we get the simple approximation

$$(1.2.3) \quad P_n\left(\frac{n}{2}\right) \sim \sqrt{\frac{2}{\pi}} \cdot \frac{1}{\sqrt{n}} \quad \text{when } p = 1/2.$$

This result confirms the conjecture that the probability tends to 0 as $n \rightarrow \infty$. The exact values of $P_n\left(\frac{n}{2}\right)$ and the approximation (1.2.3) are shown in Table 1.2.1 for varying n .

TABLE 1.2.1. $P_n\left(\frac{n}{2}\right)$ for $p = 1/2$

n	4	20	100	500	1,000	10,000
Exact	.375	.176	.0796	.0357	.0252	.00798
(1.2.3)	.399	.178	.0798	.0357	.0252	.00798

A surprising feature of the table is how slowly the probability decreases. Even for $n = 10,000$, the probability of an exactly even 5,000–5,000 split

is not much below .01. Qualitatively, this could have been predicted from (1.2.3) because of the very slow increase of $\sqrt{n}$ as a function of n . The table indicates that the approximation is highly accurate for $n > 20$.

Case 2. $p \neq 1/2$. Since $\sqrt{4pq} < 1$ for all $p \neq 1/2$ (Problem 2.2), the approximate probabilities (1.2.3) for $p = 1/2$ are multiplied by the n^{th} power of a number between 0 and 1 when $p \neq 1/2$. They are therefore greatly reduced and tend to 0 at a much faster rate. The exact values of $P_n\left(\frac{n}{2}\right)$ and the approximation (1.2.2) are shown in Table 1.2.2 for the case $p = 1/3$. Again,

TABLE 1.2.2. $P_n\left(\frac{n}{2}\right)$ for $p = 1/3$

n	4	20	100	1,000	10,000
Exact	.296	.0543	.000220	6.692×10^{-28}	1.378×10^{-258}
(2.2)	.315	.0549	.000221	6.694×10^{-28}	1.378×10^{-258}

the approximation is seen to be highly accurate for $n > 20$.

A comparison of the two tables shows the radical difference in the speed with which $P_n\left(\frac{n}{2}\right)$ tends to 0 in the two cases.

So far we have restricted attention to the probability of an even split, that is, the case in which $\frac{x}{n} = \frac{1}{2}$. Let us now consider the more general case that x/n has any given fixed value α ($0 < \alpha < 1$), which, of course, requires that αn is an integer. Then

$$P_n(x) = \binom{n}{\alpha n} (p^\alpha q^{1-\alpha})^n$$

and application of Stirling's formula shows in generalization of (1.2.2) that (Problem 2.3)

$$(1.2.4) \quad P_n(\alpha n) \sim \frac{1}{\sqrt{2\pi\alpha(1-\alpha)}} \cdot \frac{1}{\sqrt{n}} \gamma^n$$

with

$$(1.2.5) \quad \gamma = \left(\frac{p}{\alpha}\right)^\alpha \left(\frac{q}{1-\alpha}\right)^{1-\alpha}.$$

As before, there are two cases.

Case 1. $p = \alpha$. This is the case in which p is equal to the frequency of success, the probability of which is being evaluated. Here $\gamma = 1$ and (1.2.4) reduces to

$$(1.2.6) \quad P_n(\alpha n) \sim \frac{1}{\sqrt{2\pi\alpha(1-\alpha)}} \cdot \frac{1}{\sqrt{n}} \text{ when } p = \alpha.$$

TABLE 1.2.3. $P_n\left(\frac{n}{100}\right)$ for $p = .01$

$x = \frac{n}{100}$	1	2	5	10	100
n	100	200	500	1,000	10,000
Exact	.370	.272	.176	.126	.0401
(1.2.6) for $\alpha = .01$	.401	.284	.179	.127	.0401
(1.2.8)	.368	.271	.176	.125	.0399
γ	1	2	5	10	100

The exact values of $P_n(\alpha n)$ and the approximation (1.2.6) are shown in the first two rows of Table 1.2.3 for the case $\alpha = .01$. The results are similar to those of Table 1.2.1, except that the approximation (1.2.6) does not become satisfactory until $n = 500$. We shall return to this difficulty below.

Case 2. $p \neq \alpha$. Here formula (1.2.4) applies, and it can be shown that $0 < \gamma < 1$ when $p \neq \alpha$ (Problem 2.4), so that the approximate probabilities will be much smaller than when $\gamma = \alpha$. The exact values of $P_n(\alpha n)$ and the approximation (1.2.4) are shown in the first two rows of Table 1.2.4 for the case $\alpha = .01, \gamma = .02$. Again, the results are similar to those for

TABLE 1.2.4. $P_n\left(\frac{n}{100}\right)$ for $p = .02$

$x = \frac{n}{100}$	1	2	5	10	100
n	100	200	500	1,000	10,000
Exact	.271	.146	.0371	.00556	1.136×10^{-15}
(1.2.4)	.294	.152	.0377	.00560	1.137×10^{-15}
(1.2.8)	.271	.146	.0378	.00582	1.880×10^{-15}
γ	2	4	10	20	200

$\alpha = 1/2$ (Table 1.2.2) except that it requires much larger values of n before the approximation becomes satisfactory.

The diminished accuracy for $n = 100$ and 200 in Tables 1.2.3 and 1.2.4 is explained by the small values of x in these cases, since the approximations were based on the assumption that not only n but then also $x = \alpha n$ is large. The table entries in question correspond to situations in which n is large and x is small but, nevertheless, the probability $P_n(x)$ of observing x is appreciable. To obtain a better approximation for such cases, note that $P_n(x)$ is a function of the three variables x , p , and n . If p is fixed, it is nearly certain that only large values of x will occur when n is large. To obtain situations in which small values of x have appreciable probability for large n , it is necessary to allow p to become small as n gets large.

For large n , we expect x/n to be close to p ; so small values of x , say between 1 and 5, are most likely when np is in this range. This suggests letting p be proportional to $1/n$, say

$$(1.2.7) \quad p = \frac{\lambda}{n}.$$

Then

$$P_n(x) = \frac{n!}{x!(n-x)!} \left(\frac{\lambda}{n}\right)^x \left(1 - \frac{\lambda}{n}\right)^{n-x}.$$

Now

$$\frac{n!}{n^x(n-x)!} = \frac{n(n-1)\cdots(n-x+1)}{n^x} \rightarrow 1 \text{ as } n \rightarrow \infty$$

and by (1.1.8)

$$\left(1 - \frac{\lambda}{n}\right)^{n-x} \sim \left(1 - \frac{\lambda}{n}\right)^n \rightarrow e^{-\lambda},$$

so that

$$(1.2.8) \quad P_n(x) \rightarrow \frac{\lambda^x}{x!} e^{-\lambda}.$$

The terms on the right side of (1.2.8) are the probabilities $P(X = x)$ of the Poisson distribution with parameter $\lambda = E(X)$, and (1.2.8) is the Poisson approximation to the binomial distribution.

To see how well the approximation works for small x and large n , consider once more Table 1.2.4. The third row (with λ in place of γ) shows that for small values of x , the Poisson approximation is considerably more accurate than the earlier approximation (1.2.4) which was based on Stirling's formula. On the other hand, as x increases, the Poisson approximation becomes less accurate.

Formulas (1.2.4) and (1.2.8) provide two different approximations for binomial probabilities with large n . Both are obtained by considering sequences (n, p_n, x_n) for increasing n , and determining the limit behavior of $P_n(x) = f(n, x, p)$ for these sequences. A given situation can be viewed as a member of either sequence. To be specific, suppose we want an approximation for the case that $n = 100$, $x = 1$, and $p = 1/100$. We can embed this situation in the two sequences as follows.

Sequence 1: p is fixed, $x = \frac{n}{100}$, $n = 100, 200, 300, \dots$

$$n = 100, \quad x = 1, \quad p = .01$$

$$n = 200, \quad x = 2, \quad p = .01$$

$$n = 300, \quad x = 3, \quad p = .01$$

This is the sequence leading to (1.2.4) with $\alpha = .01$.

Sequence 2: x is fixed, $p = 1/n$, $n = 100, 200, 300, \dots$

$$n = 100, \quad x = 1, \quad p = .01$$

$$n = 200, \quad x = 1, \quad p = .005$$

$$n = 300, \quad x = 1, \quad p = .00333\dots$$

This is the sequence leading to (1.2.8) with $\lambda = 1$.

Table 1.2.3 shows that in this particular case, the approximation derived from the second sequence is better.

Note: The two stated sequences are of course not the only ones in which the given situation can be embedded. For another example, see Problem 2.14.

The present section illustrates a central aspect of large-sample theory: the interplay and mutual support of theoretical and numerical work. Formulas such as (1.2.4) and (1.2.8) by themselves are of only limited usefulness without some information about their accuracy such as that provided by Tables 1.2.1–1.2.4. On the other hand, the numbers shown in these tables provide only snapshots. It is only when the two kinds of information are combined that one obtains a fairly comprehensive and reliable picture.

Summary

1. For fixed p and large n , the binomial probabilities $P_n(x)$ can be well approximated (with the help of Stirling's formula) for values of x which are not too close to 0 or n by (1.2.4) and (1.2.5).
2. For small x , large n , and appreciable $P_n(x)$, a better approximation is obtained by letting $p \rightarrow 0$ in such a way that np tends to a finite limit.
3. These two examples illustrate the general method of obtaining approximations by embedding the actual situation in a sequence of situations, the limit of which furnishes the approximation. Among the many possible embedding sequences, the most useful are those that are simple and give good approximations even for relatively small n .

1.3 Infinite series

For a sequence of numbers $a_1, a_2, \dots$ consider the partial sums

$$s_n = a_1 + a_2 + \dots + a_n, \quad n = 1, 2, \dots$$

If the sequence of the sums $s_n, n = 1, 2, \dots$, tends to a limit l , we write

$$(1.3.1) \quad \sum_{i=1}^{\infty} a_i = a_1 + a_2 + \dots = l.$$

We call the left side an *infinite series* and l its sum.

Example 1.3.1 Three modes of behavior.

(i) Consider the sum

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots = \sum_{i=1}^{\infty} \frac{1}{2^i}.$$

Here the partial sums are

$$s_1 = \frac{1}{2}, \quad s_2 = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}, \quad s_3 = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} = \frac{7}{8}$$

and quite generally

$$s_n = \frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2^n}.$$

This sum is equal to (Problem 3.9)

$$\frac{1}{2^n} [1 + 2 + \dots + 2^{n-1}] = \frac{2^n - 1}{2^n} = 1 - \frac{1}{2^n}.$$

Thus $s_n \rightarrow 1$ and we can write

$$(1.3.2) \quad \sum_{i=1}^{\infty} \frac{1}{2^i} = 1.$$

(ii) If $a_i = 1/i$, the resulting series $\sum 1/i$ is the *harmonic series*. It is easily seen that

$$s_n = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} + \dots + \frac{1}{n}$$

tends to infinity by noting that

$$\frac{1}{3} + \frac{1}{4} > \frac{1}{4} + \frac{1}{4} = \frac{1}{2},$$

that

$$\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} > \frac{4}{8} = \frac{1}{2},$$

that the sum of the next 8 terms exceeds $\frac{8}{16} = \frac{1}{2}$, and so on.

A series for which s_n tends to a finite limit l is said to converge to l . A series for which $s_n \rightarrow \infty$ is said to *diverge*. A series of *positive* terms $a_1 + a_2 + \cdots$ will always tend to a limit which may be finite or infinite. It converges to a finite limit if the a 's tend to 0 sufficiently fast, as in (1.3.2). It will diverge if the a 's do not tend to 0 or if they tend to 0 sufficiently slowly, as in the harmonic series.

- (iii) If the a 's are not all positive, the series may not tend to any limit, finite or infinite. For example, if the a 's are $+1, -1, +1, -1, \dots$, then

$$s_n = \begin{cases} 1 & \text{if } n \text{ is odd} \\ 0 & \text{if } n \text{ is even} \end{cases}$$

And $\lim s_n$ does not exist. □

Example 1.3.2 Sums of powers of integers. The harmonic series of Example 1.3.1 (ii) is the special case $\alpha = 1$ of the sum of negative powers

$$(1.3.3) \quad \sum_{i=1}^{\infty} \frac{1}{i^{\alpha}}.$$

Since this series diverges for $\alpha = 1$, it must also diverge when $\alpha < 1$ since then the terms tend to zero even more slowly. In fact, it can be shown that the partial sums of (1.3.3) satisfy

$$(1.3.4) \quad s_n = \sum_{i=1}^n \frac{1}{i^{\alpha}} \sim \frac{n^{1-\alpha}}{1-\alpha} \text{ if } \alpha < 1$$

and

$$(1.3.5) \quad s_n = \sum_{i=1}^n \frac{1}{i} \sim \log n \text{ if } \alpha = 1,$$

both of which tend to ∞ as $n \rightarrow \infty$. On the other hand, when $\alpha > 1$, the terms tend to 0 fast enough for (1.3.3) to converge to a finite limit.

Formula (1.3.4) holds not only for negative but also for positive powers of the integers, so that (in generalization of (1.1.28))

$$(1.3.6) \quad s_n = \sum_{i=1}^n i^{\alpha} \sim \frac{n^{1+\alpha}}{1+\alpha} \text{ for all } \alpha > 0.$$

(For proofs of these results, see, for example, Knopp (1990).) □

Example 1.3.3 The geometric series. The series (1.3.2) is the special case of the *geometric series* $\sum x^i$ when $x = 1/2$. If we start the series at $i = 0$, we have

$$s_n = 1 + x + \cdots + x^{n-1} = \frac{1 - x^n}{1 - x}.$$

For $|x| < 1$, this tends to the limit

$$(1.3.7) \quad \sum_{i=0}^{\infty} x^i = \frac{1}{1 - x}.$$

For $x = 1$, the series diverges since $s_n = n \rightarrow \infty$. For $x = -1$, we have the case (iii) of Example 1.3.1 and s_n therefore does not tend to a limit. Similarly, the series diverges for all $x > 1$, while for $x < -1$, the series does not tend to a limit. $\square$

The geometric series is a special case of a *power series*

$$(1.3.8) \quad \sum_{i=0}^{\infty} c_i x^i.$$

In generalization of the result for the geometric series, it can be shown that there always exists a value r ($0 \leq r \leq \infty$), called the *radius of convergence* of the series, such that (1.3.8) converges for all $|x| < r$ and for no $|x| > r$; it may or may not converge for $x = \pm r$. The case $r = \infty$ corresponds to the situation that (1.3.8) converges for all x . An example is provided by the exponential series in Example 1.3.4 below. For an example with $r = 0$, see Problem 3.4.

The following are a number of important power series. For each, the sum and the radius of convergence will be stated without proof. These series will be considered further in Section 2.5.

Example 1.3.4 The exponential series. In Example 1.1.1, an expression for any power of e was given in (1.1.8). An alternative expression is provided by the infinite series

$$(1.3.9) \quad 1 + \frac{x}{1!} + \frac{x^2}{2!} + \cdots = e^x,$$

which converges for all x . For $x = \lambda > 0$, (1.3.9) shows that the Poisson distribution (1.2.8) with probabilities

$$P(X = i) = \frac{\lambda^i}{i!} e^{-\lambda}, \quad i = 0, 1, 2, \dots$$

really is a distribution, i.e., that the probabilities for $i = 0, 1, 2, \dots$ add up to 1. $\square$

Example 1.3.5 Binomial series. For any real number α , the series

$$(1.3.10) \quad 1 + \frac{\alpha}{1!}x + \frac{\alpha(\alpha-1)}{2!}x^2 + \cdots = (1+x)^\alpha$$

has radius of convergence 1.

Some special cases are worth noting.

- (i) If α is a positive integer, say $\alpha = n$, the coefficient of x^i is zero for all $i > n$ and (1.3.10) reduces to the binomial theorem

$$(1+x)^n = \sum_{i=0}^n \binom{n}{i} x^i.$$

- (ii) If α is a negative integer, say $\alpha = -m$, (1.3.10) with $x = -q$ reduces to

$$(1.3.11) \quad (1-q)^{-m} = 1 + \binom{m}{1}q + \binom{m+1}{2}q^2 + \cdots.$$

This shows that the probabilities

$$(1.3.12) \quad P(X=i) = \binom{m+i-1}{m-1} p^m q^i, \quad i = 0, 1, 2, \dots \quad (0 < p < 1, q = 1-p)$$

of the negative binomial distribution (listed in Table 1.6.2) add up to 1.

- (iii) For $m = 1$, (1.3.11) reduces to the geometric series (1.3.7) (with $x = q$) and the distribution (1.3.12) reduces to the geometric distribution with probabilities

$$(1.3.13) \quad P(X=i) = p \cdot q^i, \quad i = 0, 1, 2, \dots$$

□

Example 1.3.6 The logarithmic series. The series

$$(1.3.14) \quad x + \frac{x^2}{2} + \frac{x^3}{3} + \cdots = -\log(1-x)$$

has radius of convergence 1. When $x = 1$, it reduces to the harmonic series, which in Example 1.3.1(ii) was seen to diverge. For $0 < \theta < 1$, (1.3.14) shows that the probabilities

$$(1.3.15) \quad P(X=i) = \frac{1}{-\log(1-\theta)} \frac{\theta^i}{i}, \quad i = 1, 2, \dots$$

of the *logarithmic distribution* add up to 1. (For a discussion of this distribution, see Johnson, Kotz, and Kemp (1992), Chapter 7. □

Summary

1. An infinite series is said to converge to s if the sequence of its partial sums tends to s .
2. For any power series $\sum c_i x^i$, there exists a number $0 \leq r \leq \infty$ such that the series converges for all $|x| < r$ and for no $|x| > r$.
3. Important examples of power series are the geometric, the exponential, the binomial, and the logarithmic series.

1.4 Order relations and rates of convergence

In Section 1.1, two sequences a_n, b_n were defined to be asymptotically equivalent if $a_n/b_n \rightarrow 1$. This relation states that for large n , the numbers of the two sequences are roughly equal. In the present section, we consider three other relationships between sequences, denoted by o , $\asymp$, and O , which correspond to a_n being of smaller, equal, or less or equal, order than b_n , respectively. In a related development, we shall study the behavior of sequences exhibiting respectively slow, moderate, and rapid growth.

We begin by defining the relationship $a_n = o(b_n)$, which states that for large n , a_n is an order of magnitude smaller than b_n .

Definition 1.4.1 We say that $a_n = o(b_n)$ as $n \rightarrow \infty$ if

$$(1.4.1) \quad a_n/b_n \rightarrow 0.$$

When a_n and b_n both tend to infinity, this states that a_n tends to infinity more slowly than b_n ; when both tend to 0, it states that a_n tends to zero faster than b_n . For example,

$$(1.4.2) \quad \frac{1}{n^2} = o\left(\frac{1}{n}\right) \text{ as } n \rightarrow \infty$$

since $\frac{1}{n^2}/\frac{1}{n} = \frac{n}{n^2} \rightarrow 0$. As is shown in Table 1.4.1, it makes a big difference whether a sequence tends to 0 at rate $1/n^2$ or $1/n$.

TABLE 1.4.1. Speed of convergence of $1/n$ and $1/n^2$

n	1	2	5	10	20	100	1,000
$1/n$	1	.5	.2	.1	.05	.01	.001
$1/n^2$	1	.25	.04	.01	.0025	.0001	.000001

The following provides a statistical illustration of this difference.

Example 1.4.1 Bayes estimator for binomial p . Suppose that X has the binomial distribution $b(p, n)$ corresponding to n trials with success probability p . The standard estimator X/n is unbiased (i.e., satisfies $E(X/n) = p$) and has variance pq/n . An interesting class of competitors is the set of estimators

$$(1.4.3) \quad \delta(X) = \frac{a + X}{a + b + n}.$$

($\delta(X)$ is the Bayes estimator which minimizes the expected squared error when p has a beta prior with parameters a and b .)

Since

$$E[\delta(X)] = \frac{a + np}{a + b + n},$$

the bias of δ is

$$E[\delta(X)] - p = \frac{a + np}{a + b + n} - p,$$

which simplifies to

$$(1.4.4) \quad \text{bias of } \delta = \frac{aq - bp}{a + b + n} \text{ where } q = 1 - p.$$

Similarly, the variance of δ is

$$(1.4.5) \quad \text{Var}[\delta(X)] = \frac{npq}{(a + b + n)^2}.$$

The accuracy of an estimator $\delta(X)$ of a parameter $g(\theta)$ is most commonly measured by the expected squared error $E[\delta(X) - g(\theta)]^2$, which can be decomposed into the two terms (Problem 4.1)

$$(1.4.6) \quad E[\delta(X) - g(\theta)]^2 = (\text{bias of } \delta)^2 + \text{variance of } \delta.$$

When $\delta(X)$ is given by (1.4.3), the two components of the *bias-variance decomposition* (1.4.6) make very unequal contributions to the squared error since

$$\frac{(\text{bias of } \delta)^2}{\text{Var}[\delta(X)]} \sim \frac{(aq - bp)^2}{npq} \rightarrow 0 \text{ as } n \rightarrow \infty,$$

and hence

$$(\text{bias of } \delta)^2 = o[\text{Var } \delta(X)].$$

Both terms tend to zero, but the square of the bias tends much faster: at the rate of $1/n^2$ compared to the rate $1/n$ for the variance. The bias therefore contributes relatively little to the expected squared error. As a numerical illustration, Table 1.4.2 shows the squared bias, the variance, and the contribution of the squared bias to the total, for the case $a = b = 1/2, p = 1/3$.

TABLE 1.4.2. Squared bias and variance of a biased estimator

n	10	50	100
(1) Variance	.018	.0043	.0022
(2) (bias) ²	.00023	.000011	.0000027
(2) / [(1) + (2)]	.012	.0025	.0012

□

The results of this example are typical for standard estimation problems: The variance and bias will each be of order $1/n$, and so therefore will be the expected squared error; the relative contribution of the squared bias to the latter will tend to 0 as $n \rightarrow \infty$. An important exception is the case of nonparametric density estimation, which will be considered in Section 6.4.

The following example illustrates an important use of the o notation.

Example 1.4.2 Order of a remainder. Suppose that

$$(1.4.7) \quad a_n = \frac{1}{n} - \frac{2}{n^2} + \frac{4}{n^3}.$$

An obvious first approximation to a_n for large n is

$$(1.4.8) \quad b_n = \frac{1}{n}.$$

Clearly, $a_n \sim b_n$. Let us denote the remainder in this approximation by R_n so that

$$(1.4.9) \quad a_n = \frac{1}{n} + R_n,$$

where

$$(1.4.10) \quad R_n = \frac{-2}{n^2} + \frac{4}{n^3} = o(1/n).$$

The information provided by (1.4.9) and the right side of (1.4.10) is frequently written more compactly as

$$(1.4.11) \quad a_n = \frac{1}{n} + o\left(\frac{1}{n}\right),$$

where $o(1/n)$ denotes any quantity tending to 0 faster than $1/n$. We can refine the approximation (1.4.8) by taking into account also the $1/n^2$ term, and approximate (1.4.7) by

$$(1.4.12) \quad b'_n = \frac{1}{n} - \frac{2}{n^2};$$

we can then write

$$(1.4.13) \quad a_n = \frac{1}{n} - \frac{2}{n^2} + o\left(\frac{1}{n^2}\right).$$

Since the remainder is of order $1/n^2$ in the approximation (1.4.8) and of order $1/n^3$ in (1.4.12), we expect the second approximation to be more accurate than the first. This is illustrated by Table 1.4.3, which generally supports this expectation but shows that for very small n , the first approximation is, in fact, closer in this case. $\square$

TABLE 1.4.3. The accuracy of two approximations

n	1	2	3	5	10	50
$\frac{1}{n} - \frac{2}{n^2} + \frac{4}{n^3}$	3.0	.5	.259	.152	.84	.0192
$1/n$	1.0	.5	.333	.200	1.00	.0200
$\frac{1}{n} - \frac{2}{n^2}$	-1.0	0.0	.111	.120	.80	.0197

Note: The approximation (1.4.8) which takes account only of the principal term ($1/n$) of (1.4.7) is a *first order approximation* while (1.4.12), which also includes the next ($1/n^2$) term, is of *second order*.

While higher order approximations tend to be more accurate, they also suffer from some disadvantages: They are more complicated and, for the probability calculations to be considered later, they typically require more knowledge of the mathematical model. In addition, the higher order theory is more difficult mathematically. For these reasons, in this book we shall restrict attention mainly to first order approximations and shall make only a few brief excursions into higher order territory.

The following are some simple properties of the o relation (Problem 4.3).

Lemma 1.4.1

(i) A sequence of a 's satisfies

$$(1.4.14) \quad a_n = o(1) \text{ if and only if } a_n \rightarrow 0.$$

(ii) If $a_n = o(b_n)$ and $b_n = o(c_n)$, then $a_n = o(c_n)$.

(iii) For any constant $c \neq 0$,

$$(1.4.15) \quad a_n = o(b_n) \text{ implies } ca_n = o(b_n).$$

(iv) For any sequence of number c_n different from 0,

$$a_n = o(b_n) \text{ implies } c_n a_n = o(c_n b_n).$$

(v) If $d_n = o(b_n)$ and $e_n = o(c_n)$, then $d_n e_n = o(b_n c_n)$.

Example 1.4.3 The order relationship of powers of n . If $a_n = n^\alpha$, $b_n = n^\beta$, then $a_n = o(b_n)$, provided $\alpha < \beta$. To see this, note that the ratio

$$\frac{a_n}{b_n} = \frac{n^\alpha}{n^\beta} = \frac{1}{n^{\beta-\alpha}}$$

tends to zero when $\alpha < \beta$. Thus $\sqrt[4]{n}, \sqrt{n}, n, n^2, n^3, \dots$ constitute sequences which tend to infinity as $n \rightarrow \infty$ with increasing rapidity. $\square$

Example 1.4.4 Exponential growth. For any $k > 0$ and $a > 1$, we have

$$(1.4.16) \quad n^k = o(a^n).$$

To prove (1.4.16), write $a = 1 + \epsilon$ ($\epsilon > 0$), and consider the case $k = 2$. Instead of showing that $n^2/a^n \rightarrow 0$, we shall prove the equivalent statement that

$$a^n/n^2 \rightarrow \infty.$$

For this purpose, we use the expansion

$$a^n = (1 + \epsilon)^n = 1 + n\epsilon + \binom{n}{2}\epsilon^2 + \binom{n}{3}\epsilon^3 + \dots > \binom{n}{3}\epsilon^3,$$

which shows that

$$\frac{a^n}{n^2} > \frac{n(n-1)(n-2)}{n^2} \cdot \frac{\epsilon^3}{6} = \left(n-3 + \frac{2}{n}\right) \cdot \frac{\epsilon^3}{6}.$$

Since the right side tends to ∞ as $n \rightarrow \infty$, this completes the proof for the case $k = 2$. The argument for any other integer k is quite analogous. If k is not an integer, let k' be any integer greater than k . Then

$$n^k = o(n^{k'}) \text{ and } n^{k'} = o(a^n),$$

and the result follows from part (ii) of Lemma 1.4.1.

More generally, it is seen that (Problem 4.7)

$$(1.4.17) \quad c_0 + c_1 n + \dots + c_k n^k = o(a^n)$$

for any integer k , any $a > 1$, and any constants $c_0, c_1, \dots, c_k$. Sequences that are asymptotically equivalent either to (1.4.17) or to a^n ($a > 1$) are

said to grow at polynomial or exponential rate, respectively. Exponential growth is enormously much faster than polynomial growth, as is illustrated in the first two rows of Table 1.4.4. This explains the surprising growth due to compound interest (Problem 4.9).

Note: Because of this very great difference, it has been an important issue in computer science to determine whether certain tasks of size n require exponential time (i.e., a number of steps that increases exponentially) or can be solved in polynomial time. $\square$

Let us next consider some sequences that increase more slowly than n^k for any $k > 0$.

Example 1.4.5 Logarithmic growth. Let $a_n = \log n, b_n = n^k$. Then $a_n = o(b_n)$ for every $k > 0$. To show that $\frac{\log n}{n^k} \rightarrow 0$, it is enough by Definition 1.1.1 to show that for any $\epsilon > 0$, we have $-\epsilon < \frac{\log n}{n^k} < \epsilon$ for all sufficiently large n . Since $\log n/n^k$ is positive, it is therefore enough to show that for all sufficiently large n

$$n < e^{\epsilon n^k},$$

which follows from Problem 4.7(ii). $\square$

The three scales a^n , n^k , and $\log n$ are the most commonly used scales for the growth of a sequence tending to infinity, representing fast, moderate, and slow growth, respectively. (Correspondingly their reciprocals a^{-n} , n^{-k} , and $1/\log n$ provide standard scales for convergence to 0.) One can get an idea of the differences in these rates by looking at the effect of doubling n . Then, for example,

$$a_n = a^n \text{ becomes } a_{2n} = a^{2n} = a^n a^n,$$

$$a_n = n^2 \text{ becomes } a_{2n} = 4n^2 = 4a_n,$$

$$a_n = \log n \text{ becomes } a_{2n} = \log(2n) = a_n + \log 2.$$

Thus doubling n multiplies a_n in the first case by a factor which rapidly tends to ∞ ; in the second case, by the fixed factor 4; finally in the third case, doubling n only adds the fixed amount $\log 2$ to a_n . Table 1.4.4 illustrates these three rates numerically. While the first sequence seems to explode, the last one barely creeps along; yet eventually it too will become arbitrarily large (Problem 4.10(i)).

If a_n/b_n tends to 0 or ∞ , a_n is respectively of smaller or larger order than b_n . In the intermediate case that a_n/b_n tends to a finite non-zero limit, the two sequences are said to be of the same order or to tend to 0 or ∞ at the same rate. (This terminology was already used informally at the end of

TABLE 1.4.4. Three rates of invergence

n	1	2	5	10	15	20
e^n	2.7	7.4	148.4	22,026	3,269,017	4,851,165,195
n^3	1	8	125	1000	3375	8000
$\log n$	0	.7	1.6	2.3	2.7	3.0

Example 1.4.1 when it was stated that the bias and variance are typically “of order $1/n$.”)

The concept of two sequences being of the same order can be defined somewhat more generally without requiring a_n/b_n to tend to a limit.

Definition 1.4.2 Two sequences $\{a_n\}$ and $\{b_n\}$ are said to be of the same order, in symbols,

$$(1.4.18) \quad a_n \asymp b_n,$$

if $|a_n/b_n|$ is bounded away from both 0 and ∞ , i.e., if there exist constants $0 < m < M < \infty$ and an integer n_0 such that

$$(1.4.19) \quad m < \left| \frac{a_n}{b_n} \right| < M \text{ for all } n > n_0.$$

As an example, let

$$a_n = 2n + 3, \quad b_n = n.$$

Then

$$\frac{a_n}{b_n} < 3 \text{ holds when } 2n + 3 < 3n,$$

i.e., for all $n > 3$, and

$$\frac{a_n}{b_n} > 2 \text{ holds for all } n.$$

Thus (1.4.19) is satisfied with $m = 2$, $M = 3$, and $n_0 = 3$.

It is easy to see that if a_n/b_n tends to any finite non-zero limit, then $a_n \asymp b_n$ (Problem 4.11).

Lemma 1.4.2 If $a_n \asymp b_n$, then $ca_n \asymp b_n$ for any $c \neq 0$.

The symbol $\asymp$ denotes equality of the orders of magnitude of a_n and b_n and constitutes a much cruder comparison than the asymptotic equivalence $a_n \sim b_n$. This is illustrated by the statement of Lemma 1.4.2 that the relationship $a_n \asymp b_n$ is not invalidated when a_n is multiplied by any fixed constant $c \neq 0$. Definition 1.4.2 of equality of order is motivated by

the extreme difference in the speed of convergence of sequences of different order shown in Table 1.4.4. In the light of such differences, multiplication by a fixed constant does little to change the basic picture.

In addition to o and $\asymp$, it is useful to have a notation also for a_n being of order smaller than or equal to that of b_n . This relationship is denoted by $a_n = O(b_n)$. It follows from Definitions 1.4.1 and 1.4.2 that $a_n = O(b_n)$ if $|a_n/b_n|$ is bounded, i.e., if there exist M and n_0 such that

$$(1.4.20) \quad \left| \frac{a_n}{b_n} \right| < M \text{ for all } n > n_0.$$

Note in particular that

$$(1.4.21) \quad a_n = O(1) \text{ if and only if the sequence } \{a_n\} \text{ is bounded.}$$

Example 1.4.6 Information provided by $\sim$, $\asymp$, O , and o . To illustrate the distinction among $\sim$, $\asymp$, O , and o and the information they provide about a remainder, suppose that

$$a_n = \frac{1}{n} + \frac{b}{n\sqrt{n}} + \frac{c}{n^2} + \frac{d}{n^2\sqrt{n}}$$

is approximated by $1/n$, so that

$$a_n = \frac{1}{n} + R_n$$

with

$$R_n = \frac{b}{n\sqrt{n}} + \frac{c}{n^2} + \frac{d}{n^2\sqrt{n}}.$$

Then the following implications hold (Problem 4.20):

- (i) $R_n \sim \frac{1}{n^2}$ if and only if $b = 0, c = 1$;
- (ii) $R_n \asymp \frac{1}{n^2}$ if and only if $b = 0, c \neq 0$;
- (iii) $R_n = O\left(\frac{1}{n^2}\right)$ if and only if $b = 0$;
- (iv) $R_n = o\left(\frac{1}{n^2}\right)$ if and only if $b = c = 0$.

□

Summary

1. For two sequences $\{a_n\}, \{b_n\}$, we consider the possibilities that

- (a) a_n is of smaller order than b_n ; in symbols, $a_n = o(b_n)$,
- (b) a_n is of the same order as b_n ; in symbols, $a_n \asymp b_n$,
- (c) a_n is of order smaller than or equal to b_n ; in symbols, $a_n = O(b_n)$,

and we discuss some properties of these order relations.

- 2. Three of the most commonly encountered growth rates are the logarithmic ($\log n$), polynomial (n^k), and exponential ($a^n, a > 1$), and they satisfy

$$\log n = o(n^k) \text{ for all } k > 0,$$

$$n^k = o(a^n) \text{ for all } k > 0, a > 1.$$

- 3. When approximating a complicated a_n by a simpler b_n with remainder R_n , the three orders described in 1 can be used to characterize our knowledge about the order of the remainder.

1.5 Continuity

So far we have been concerned with the behavior of sequences $a_n, n = 1, 2, \dots$. If the notation is changed from a_n to $a(n)$, it becomes clear that such a sequence is a function defined over the positive integers. In the present section, some of the earlier ideas will be extended to functions of a continuous variable x . Such a function is a rule or formula which assigns to each value of x a real number $f(x)$, for example, $x^2 + 1$, e^x , or $\sin x$. The function will be denoted by f , the value that f assigns to x by $f(x)$.

To define the concept of limit, we begin with the limit of $f(x)$ as $x \rightarrow \infty$, which is completely analogous to the Definition 1.1.1 of the limit of a sequence.

Definition 1.5.1 As $x \rightarrow \infty$, $f(x)$ tends (or converges) to a ; in symbols,

$$(1.5.1) \quad f(x) \rightarrow a \text{ as } x \rightarrow \infty \text{ or } \lim_{x \rightarrow \infty} f(x) = a,$$

if, given any $\epsilon > 0$, there exists $M = M(\epsilon)$ such that

$$|f(x) - a| < \epsilon \text{ for all } x > M.$$

A new situation arises when x tends to a finite value x_0 , because it can then tend to x_0 from the right (i.e., through values $> x_0$) or from the left (i.e., through values $< x_0$). We shall denote these two possibilities by $x \rightarrow x_0+$ and $x \rightarrow x_0-$, respectively.

We shall say that $f(x) \rightarrow a$ as $x \rightarrow x_0+$ if $f(x)$ gets arbitrarily close to a as x gets sufficiently close to x_0 from the right. This is formalized in

Definition 1.5.2 As $x \rightarrow x_0+$, $f(x)$ tends to a ; in symbols,

$$(1.5.2) \quad f(x) \rightarrow a \text{ as } x \rightarrow x_0+ \text{ or } \lim_{x \rightarrow x_0+} f(x) = a,$$

if, given any $\epsilon > 0$, there exists $\delta > 0$ such that

$$(1.5.3) \quad |f(x) - a| < \epsilon \text{ for all } x_0 < x < x_0 + \delta.$$

The limit of $f(x)$ as $x \rightarrow x_0-$ is defined analogously.

Example 1.5.1 Let

$$(1.5.4) \quad f(x) = \begin{cases} x^3 + 1 & \text{if } x > 0 \\ x^3 - 1 & \text{if } x < 0. \end{cases}$$

Then

$$(1.5.5) \quad \begin{aligned} f(x) &\rightarrow 1 && \text{as } x \text{ tends to } 0 \text{ from the right,} \\ f(x) &\rightarrow -1 && \text{as } x \text{ tends to } 0 \text{ from the left.} \end{aligned}$$

To see the first of these statements, note that for $x > 0$,

$$|f(x) - 1| = x^3 < \epsilon \text{ when } 0 < x < \sqrt[3]{\epsilon},$$

so that (1.5.3) holds with $\delta = \sqrt[3]{\epsilon}$. □

The one-sided limits $f(x_0+)$ and $f(x_0-)$ need not exist as is shown by the classical example

$$(1.5.6) \quad f(x) = \sin \frac{1}{x}$$

Here there exist positive values of x arbitrarily close to 0 for which $f(x)$ takes on all values between -1 and $+1$ (Problem 5.3) so that $f(x)$ does not tend to a limit as $x \rightarrow 0+$, and the same is true as $x \rightarrow 0-$.

Two-sided limits are obtained by combining the two one-sided concepts.

Definition 1.5.3 We say that

$$f(x) \rightarrow a \text{ as } x \rightarrow x_0$$

if

$$(1.5.7) \quad \lim_{x \rightarrow x_0+} f(x) = \lim_{x \rightarrow x_0-} f(x) = a,$$

i.e., if given any $\epsilon > 0$, there exists $\delta > 0$ such that

$$(1.5.8) \quad |f(x) - a| < \epsilon \text{ for all } x \neq x_0 \text{ satisfying } |x - x_0| < \delta.$$

In Example 1.5.1, the limit as $x \rightarrow 0$ does not exist since the two one-sided limits are not equal. On the other hand, if $f(x) = x^3$, it is seen that $f(x) \rightarrow 0$ as $x \rightarrow 0+$ and as $x \rightarrow 0-$, and that, therefore, $\lim_{x \rightarrow 0} f(x)$ exists and is equal to 0.

It is interesting to note that the definitions of $\lim_{x \rightarrow x_0+} f(x)$, $\lim_{x \rightarrow x_0-} f(x)$, and hence $\lim_{x \rightarrow x_0} f(x)$ say nothing about the value $f(x_0)$ that f takes on at the point x_0 itself but involves only values of f for x close to x_0 .

The relations $\sim$, $\asymp$, O , and o extend in the obvious way. For example,

$$f(x) = o[g(x)] \text{ as } x \rightarrow x_0 \quad (x_0 \text{ finite or infinite})$$

if $f(x)/g(x) \rightarrow 0$ as $x \rightarrow x_0$.

The limit laws (1.1.10)–(1.1.12) also have their obvious analogs; for example, if $f(x) \rightarrow a$ and $g(x) \rightarrow b$ as $x \rightarrow x_0$, then $f(x) + g(x) \rightarrow a + b$. Similarly, the results of Examples 1.4.3 to 1.4.5 continue to hold when n is replaced by x ; for example,

$$(1.5.9) \quad \frac{\log x}{x^\alpha} \rightarrow 0 \text{ for any } \alpha > 0 \text{ as } x \rightarrow \infty.$$

Definition 1.5.4 A function f is said to be continuous at x_0 if $f(x_0+)$ and $f(x_0-)$ exist and if

$$(1.5.10) \quad f(x_0+) = f(x_0-) = f(x_0).$$

Any polynomial $P(x)$ is continuous for all x ; so is any rational function, i.e., the ratio $P(x)/Q(x)$ of two polynomials, at any point x for which $Q(x) \neq 0$. The exponential function e^x is continuous for all x , and $\log x$ for all $x > 0$ (the latter is not defined for $x \leq 0$).

Continuity can be violated in a number of ways:

- (a) Even if $f(x_0+)$ and $f(x_0-)$ exist and are equal, $f(x_0+) = f(x_0-) = a$, say, it may happen that $f(x_0) \neq a$.

As an example, let

$$f(x) = \begin{cases} x^2 & \text{if } x \neq 2 \\ 0 & \text{if } x = 2. \end{cases}$$

- (b) A more interesting possibility is that $f(x_0+) \neq f(x_0-)$, as was the case in Example 1.5.1.

Definition 1.5.5 The function f is said to be continuous on the right at x_0 if $f(x_0+)$ exists and

$$(1.5.11) \quad f(x_0) = f(x_0+).$$

It is continuous on the left at x_0 if $f(x_0-)$ exists and

$$(1.5.12) \quad f(x_0) = f(x_0-).$$

If $f(x_0+)$ and $f(x_0-)$ exist, whether f is continuous on the right, on the left, or neither depends on the value $f(x_0)$. The function f given by (1.5.4) is continuous on the right at 0 if $f(0) = 1$ and continuous on the left if $f(0) = -1$. If $f(0)$ has a value different from ± 1 , it is continuous neither on the right nor left at 0.

A function that is continuous on both the left and right at x_0 is continuous at 0.

Discontinuities of both types (a) and (b) are called *simple discontinuities* or *discontinuities of the first kind*.

A classical example of the third possibility in which $f(x_0+)$ and/or $f(x_0-)$ do not exist is provided by the function (1.5.6).

As this example shows, discontinuities that are not simple can occur. They can, however, be ruled out for the important class of monotone functions, i.e., functions that are either non-decreasing or non-increasing. This follows from the following theorem which for the sake of simplicity is stated for the non-decreasing case.

Theorem 1.5.1

(i) If a_n is a non-decreasing sequence (i.e., satisfies $a_n \leq a_{n+1}$ for all n), then either

(a) a_n is bounded above and tends to a finite limit as $n \rightarrow \infty$

or

(b) $a_n \rightarrow \infty$ as $n \rightarrow \infty$.

(ii) Let f be a function defined for all $a < x < b$ ($-\infty \leq a < b \leq \infty$) which is non-decreasing (i.e., satisfies $f(x) \leq f(y)$ when $x < y$). If $a \leq x_0 \leq b$, then as x tends to x_0 from the left (or the right), $f(x)$ tends either to a finite limit or to ∞ ($-\infty$).

The proof (see Problem 5.11) depends on the following basic property of real numbers, discussed, for example, in Rudin (1976) and Hardy (1908, 1992): Let S be a set of real numbers which is bounded above, i.e., there exists a constant M such that $x \leq M$ for all x in S . Then among all the upper bounds M , there exists a smallest one, which will be denoted by $\sup S$. Analogously, if S is bounded below, there exists a largest lower bound, denoted by $\inf S$.

The least upper or greatest lower bound of a set S is a member of S , or a limit of points in S , or both (Problem 5.12).

Example 1.5.2 Let S be the set of numbers $1 - \frac{1}{n}$, $n = 1, 2, 3, \dots$. Then $\inf S = 0$ and $\sup S = 1$. The first of these is a member of S ; the second is not. □

Summary

1. The definition of a limit of a sequence of numbers is extended to one- and two-sided limits of a continuous variable.
2. A function f is *continuous* at a point x_0 if its limit as $x \rightarrow x_0$ exists and is equal to $f(x_0)$.
3. A function f has a simple discontinuity at x_0 if the limits $f(x_0-)$ and $f(x_0+)$ as x tends to x_0 from the left or right exist but are either not equal or are equal but $\neq f(x_0)$.
4. If a function f is monotone, the limits $f(x_0-)$ and $f(x_0+)$ exist for all x_0 , and f can therefore have only simple discontinuities.

1.6 Distributions

The principal properties of random variables are reflected in their distributions and the principal aspects of the limiting behavior of a sequence of random variables in the limiting behavior of their distributions. Probability distributions can be represented in a number of different ways, and we shall in the present section consider several such representations, particularly cumulative distribution functions, probability densities, and quantile functions.

Definition 1.6.1 The cumulative distribution function (cdf) of a random variable X is defined as

$$(1.6.1) \quad F(x) = P(X \leq x).$$

Cumulative distribution functions have the following properties.

Theorem 1.6.1

- (i) $0 \leq F(x) \leq 1$ for all x ;
- (ii) F is non-decreasing;
- (iii) $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow \infty} F(x) = 1$.
- (iv) All discontinuities of F are simple, that is, $F(x-)$ and $F(x+)$ exist for all x .
- (v) F is continuous on the right, that is, $F(x) = F(x+)$ for all x .
- (vi) The jump of $F(x)$ at a discontinuity point x is equal to

$$(1.6.2) \quad F(x) - F(x-) = P(X = x).$$

Proof. Parts (i) and (ii) are obvious; part (iv) follows from (ii) and Theorem 5.1. For proofs of the remaining parts, see, for example, Hoel, Port, and Stone (1971) or Parzen (1960, 1992). ■

It follows from (vi) that F is continuous at x if and only if

$$P(X = x) = 0.$$

Such a point x is called a *continuity point* of F .

In this book, we shall be concerned primarily with two types of distributions:

- (i) Continuous distributions F which, in addition, are differentiable, that is, possess a probability density f satisfying

$$(1.6.3) \quad F(x) = \int_{-\infty}^x f(t) dt.$$

- (ii) Discrete distributions, that is, distributions of a random variable X for which there exists a countable set A of values $a_1, a_2, \dots$ such that

$$(1.6.4) \quad \sum_{i=1}^{\infty} P(X = a_i) = 1.$$

TABLE 1.6.1. Some common probability densities

Density	Name	Notation
$\frac{1}{\sqrt{2\pi b}} e^{-(x-a)^2/b^2}$	Normal	$N(a, b^2)$
$\frac{1}{b} \frac{e^{(x-a)/b}}{[1 + e^{(x-a)/b}]^2}$	Logistic	$L(a, b)$
$\frac{1}{b} e^{-(x-a)/b}$ if $x > a$	Exponential	$E(a, b)$
$\frac{1}{2b} e^{- x-a /b}$	Double Exponential	$DE(a, b)$
$\frac{1}{b}$ if $a - \frac{b}{2} < x < a + \frac{b}{2}$	Uniform	$U\left(a - \frac{b}{2}, a + \frac{b}{2}\right)$
$\frac{b}{\pi} \frac{1}{b^2 + (x-a)^2}$	Cauchy	$C(a, b)$

Examples of the continuous case are the normal, exponential, χ^2 , uniform, and Cauchy distributions. Discrete distributions of particular importance are *lattice distributions*, i.e., distributions for which the set A is the

set $\{a + k\Delta, k = 0, \pm 1, \pm 2, \dots\}$ for some a and some $\Delta > 0$. Examples of lattice distributions are the binomial and Poisson distributions. Distributions that are continuous but not differentiable or that are partly continuous and partly discrete are of lesser interest, although they are included in some results concerning general classes of distributions.

The following result states some distributional facts that will often be used.

Theorem 1.6.2 *Let X_i ($i = 1, \dots, n$) be independently distributed according to F_i . Then in the notation of Tables 1.6.1 and 1.6.2:*

- (i) *If F_i is the Poisson distribution $P(\lambda_i)$, $Y = \sum X_i$ is distributed according to $P(\sum \lambda_i)$.*
- (ii) *If F_i is the normal distribution $N(\xi_i, \sigma_i^2)$, $Y = \sum X_i$ is distributed according to $N(\sum \xi_i, \sum \sigma_i^2)$.*
- (iii) *If F_i is the Cauchy distribution $C(a_i, b_i)$, $Y = \sum X_i$ is distributed according to $C(\sum a_i, \sum b_i)$.*

So far we have represented a probability distribution by its cdf (1.6.1); alternatively, we might have used

$$(1.6.5) \quad F^*(x) = P(X < x).$$

This alternative satisfies (i)–(iv) and (vi) of Theorem 1.6.1, but condition (v), continuity on the right, is replaced by continuity on the left.

In the continuous case (1.6.3), a distribution is often represented by its density f ; in the discrete case, by its probability function

$$(1.6.6) \quad P(X = a_i), \quad i = 1, 2, \dots$$

Tables 1.6.1 and 1.6.2 list some of the more common densities and probability functions.

Still another representation is by means of the inverse F^{-1} of F , the so-called *quantile* function.

- (a) If F is continuous and strictly increasing, F^{-1} is defined by

$$(1.6.7) \quad F^{-1}(y) = x \text{ when } y = F(x).$$

- (b) If F has a discontinuity at x_0 , suppose that $F(x_0-) < y < F(x_0) = F(x_0+)$. In this case, although there exists no x for which $y = F(x)$, $F^{-1}(y)$ is defined to be equal to x_0 .

- (c) Condition (1.6.7) becomes ambiguous when F is not strictly increasing. Suppose that

$$(1.6.8) \quad F(x) \begin{cases} < y & \text{for } x < a \\ = y & \text{for } a \leq x \leq b \\ > y & \text{for } x > b. \end{cases}$$

TABLE 1.6.2. Some common discrete distributions

Probability function $P(X = x)$	Values	Name	Notation
$\binom{n}{x} p^x (1-p)^{n-x}$	$x = 0, 1, \dots, n$	Binomial	$b(p, n)$
$\frac{1}{x!} \lambda^x e^{-\lambda}$	$x = 0, 1, \dots$	Poisson	$P(\lambda)$
$\binom{m+x-1}{m-1} p^m q^x$	$x = 0, 1, \dots$	Negative Binomial	$Nb(p, m)$
$\frac{\binom{D}{x} \binom{N-D}{n-x}}{\binom{N}{n}}$	$\max(0, n - (N - D)) \leq x \leq \min(n, D)$	Hypergeometric	$H(N, D, n)$

Then any value $a \leq x \leq b$ could be chosen for $x = F^{-1}(y)$. The convention in this case is to define

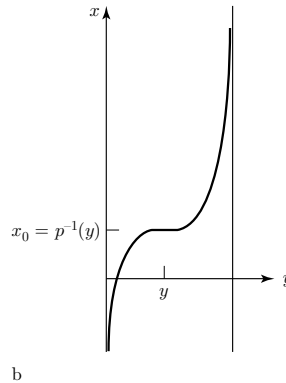
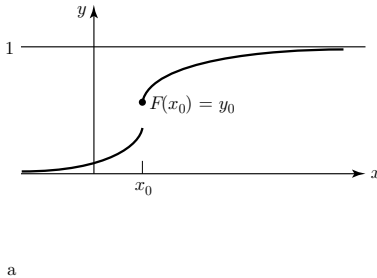
$$(1.6.9) \quad F^{-1}(y) = a,$$

i.e., as the smallest value x for which $F(x) = y$. This causes F^{-1} to be continuous on the left.

The three cases can be combined into the single definition

$$(1.6.10) \quad F^{-1}(y) = \min \{x: F(x) \geq y\}.$$

With these conventions, the graph of F^{-1} is obtained by rotating the graph of $y = F(x)$ around the line $y = x$ as the axis, so that the x -axis points up and the y -axis to the right. (A simple way of accomplishing this is to graph $y = F(x)$ on semi-transparent paper and then to reverse the sheet to obtain the graph of $x = F^{-1}(y)$.) The resulting picture, illustrated in Figure 1.6.1, shows that F^{-1} is defined on $(0, 1)$, and that discontinuities of F become converted into flat stretches of F^{-1} and flat stretches of F into discontinuities of F^{-1} .

FIGURE 1.6.1(a): $y = F(x)$ FIGURE 1.6.1(b): $x = F^{-1}(y)$

The term *quantile function* reflects the close connection of F^{-1} with the quantiles of F . In case (a), the p -quantile of F is defined as the point x such that the probability $F(x)$ to the left of x is equal to p , that is, as $x = F^{-1}(p)$. In case (b), when $p = y$, the p -quantile is the unique point x for which $F(x-) < p < F(x)$.

Note: The quantiles corresponding to $p = 1/2$, $p = 1/4$, and $p = 3/4$ are called the *median*, *first quartile*, and *third quartile*, respectively.

In case (c), there is no uniquely agreed upon definition of quantile. If (1.6.8) holds, some authors call any x in the interval $[a, b]$ a y -quantile, some define the y -quantile uniquely as a , while at least in the case of the median, still others define it to be $\frac{1}{2}(a + b)$, the midpoint of the interval $[a, b]$.

Cumulative distribution functions, probability densities, and quantile functions constitute three possible ways of representing the distribution of a random variable. Still other representations (which we shall not consider here) are by means of moment generating functions or characteristic functions.

Summary

1. The distributions of a random variable can be represented in various ways: for example, by its cumulative distribution function, by its probability density function, and by its quantile function.
2. Of particular interest are (i) the continuous case in which the cdf is differentiable, its derivative being the probability density; and (ii) the discrete case. Two tables exhibit some of the most common densities in case (i) and probability functions in case (ii).
3. Cumulative distribution functions are non-decreasing and hence any of their discontinuities must be simple. The points of discontinuity of the cdf of a random variable X are the points x for which $P(X = x)$ is positive and this probability then equals the size of the jump at x .

1.7 Problems

Section 1

- 1.1 (i) Determine $n_0(\epsilon)$ of Definition 1.1.1 when a_n is given by (1.1.2).
(ii) Compare the smallest possible values of $n_0(\epsilon)$ for the cases that a_n is given by (1.1.1) or (1.1.2) when $\epsilon = 1/1,000,000$.

- 1.2 (i) Show that the sequence a_n defined by (1.1.7) is also given by the single formula

$$a_n = 1 + (-1)^n \cdot \frac{1}{n}.$$

- (ii) If a_n is the n^{th} member of the sequence (1.1.9), find a formula for a_n analogous to that of (i).

- 1.3 Determine $n_0(M)$ of Definition 1.1.2 when

(a) $a_n = \sqrt{n}$,

(b) $a_n = n^2$,

and compare their values for the case $M = 10^4$.

- 1.4 Define the limit $a_n \rightarrow -\infty$ in analogy with Definition 1.1.2.

- 1.5 If $f(a) = c_0 + c_1a + \cdots + c_k a^k$ for all a , use (1.1.10) and (1.1.11) to show that $f(a_n) \rightarrow f(a_0)$ when $a_n \rightarrow a_0$.

- 1.6 If $a_n \rightarrow \infty$, then

(i) $\sqrt{a_n} \rightarrow \infty$;

(ii) $\log a_n \rightarrow \infty$;

(iii) $1/a_n \rightarrow 0$.

- 1.7 (i) Show that the sequence (1.1.9) tends to 0 as $n \rightarrow \infty$.

- (ii) Make a table of a_n given by (1.1.9) for $n = 75, 76, \dots, 100$.

- 1.8 Let $a_n \rightarrow \infty$ and let $b_n = a_{n+1}$. Determine whether $a_n \sim b_n$ for the following cases:

(i) $a_n = n^k$, k is a positive integer;

(ii) $a_n = e^{\alpha n}$, $\alpha > 0$;

(iii) $a_n = \log n$.

[**Hint (iii):** Apply Lemma 1.1.1 with $b_n = a_{n+1}$.]

- 1.9 Replace each of the following quantities by a simpler one that is asymptotically equivalent as $n \rightarrow \infty$:

(i) $n + \frac{1}{2}n^2 + \frac{1}{3}n^3$;

(ii) $n + 2 + \frac{3}{n}$;

(iii) $\frac{1}{n} + \frac{2}{n^2} + \frac{3}{n^3}$.

1.10 (i) If $a_n \sim b_n$, $b_n \sim c_n$, then $a_n \sim c_n$;

(ii) if $a_n \sim b_n$, then $\frac{1}{a_n} \sim \frac{1}{b_n}$.

(iii) If $a_n \sim b_n$, $c_n \sim d_n$, then $a_n c_n \sim b_n d_n$.

(iv) If $a_n \sim b_n$, then $a_n^\alpha \sim b_n^\alpha$ for any $\alpha > 0$.

1.11 If $a_n \sim b_n$, $c_n \sim d_n$, then

(i) $a_n + c_n \sim b_n + d_n$, provided a_n , b_n , c_n , and d_n are all > 0 ;

(ii) The result of (i) need not hold without the restriction to positive values.

1.12 If $a_n \rightarrow \infty$, determine whether $a_n \sim b_n$ implies that $a_n^2 + a_n \sim b_n^2 + b_n$.

[**Hint:** $a_n^2 + a_n = a_n(a_n + 1)$.]

1.13 Calculate the absolute and relative error in (1.1.26) for $n = 7$.

1.14 (i) Determine the sum $\sum_{i=1}^n i^3$ and check that (1.1.28) holds for $k = 3$.

(ii) Prove (1.1.28) for general k .

[**Hint:** (i) Add both sides of the equation $(i+1)^4 - i^4 = 4i^3 + 6i^2 + 4i + 1$ from $i = 1$ to $i = n + 1$. (ii) Generalize the method of (i) and use induction.]

1.15 If k remains fixed and $N \rightarrow \infty$, use Stirling's formula to show that

$$(1.7.1) \quad \binom{N}{k} \sim \frac{1}{k!} N^k$$

and make a table giving an idea of the accuracy of this approximation.

1.16 (i) For fixed α ($0 < \alpha < 1$), if $k = \alpha N$ is an integer and $N \rightarrow \infty$, show that

$$\binom{N}{k} \sim \frac{1}{\sqrt{2\pi\alpha(1-\alpha)N}} \cdot \frac{1}{\alpha^k (1-\alpha)^{N-k}}.$$

(ii) Compare this approximation with that of the preceding problem for some representative values of k and N .

1.17 If $0 < \gamma < 1$, then $a_n = \gamma^n \rightarrow \infty$.

[**Hint:** Since $\gamma > 0$, it is enough to show that for any $\epsilon > 0$ there exists n_0 such that $\gamma^n < \epsilon$ for all $n > n_0$. Take logarithms on both sides of this last inequality.]

1.18 (i) For any positive γ ,

$$(1.7.2) \quad a_n = \sqrt[n]{\gamma} \rightarrow 1 \text{ as } n \rightarrow \infty.$$

(ii) For $\gamma = 1/8, 1/4, 1/2, 2, 4, 8$, make a table showing the convergence of a_n to 1.

[**Hint:** Suppose $\gamma < 1$. Then it is enough to show that for any ϵ , there exists n_0 such that $\sqrt[n]{\gamma} < 1 + \epsilon$; take logarithms. The case $\gamma < 1$ can be treated analogously.]

1.19 Make a table showing the absolute and relative error resulting from approximating $S_n^{(2)}$ (defined by (1.1.27)) by $n^3/3$.

Section 2

2.1 Verify formula (1.2.2).

2.2 Show that $\sqrt{4pq} < 1$ for all $p \neq \frac{1}{2}$ ($0 \leq p \leq 1$).

2.3 Verify formula (1.2.4).

2.4 Show that γ defined by (1.2.5) is < 1 for $p \neq \alpha$.

2.5 For fixed x as $n \rightarrow \infty$, the probability (1.2.1) satisfies

$$P_n(x) \sim n^x p^x q^{n-x} = \left(\frac{np}{q}\right)^x q^n.$$

2.6 Let X be distributed according to the hypergeometric distribution shown in Table 1.6.2. If n is held fixed and D and N tend to infinity in such a way that $\frac{D}{N} = p$ remains fixed, use (1.7.1) to show that

$$(1.7.3) \quad P(X = x) \rightarrow \binom{n}{x} p^x q^{n-x}.$$

2.7 Make tables similar to Table 1.2.2 for $p = 1/4$ and $p = 1/10$.

2.8 Let X be binomial $b(p, n)$ with $p = 1/2$ and n even. Generalize formula (1.2.3) to determine whether for fixed d

$$(1.7.4) \quad P\left(X = \frac{n}{2} + d\right) \sim \frac{c}{\sqrt{n}}$$

continues to hold for some c .

2.9 For $x = 0$ and $x = 1$, make a table showing the accuracy of the Poisson approximation (1.2.8) for $\lambda = .1, .5, 1$ and $n = 5, 10, 20, 50, 100$.

2.10 Let X have the Poisson distribution $P(\lambda)$ shown in Table 1.6.2.

(i) Show that for $x = \alpha\lambda$ (α fixed, $\lambda \rightarrow \infty$),

$$(1.7.5) \quad P(X = x) \sim \frac{1}{\sqrt{2\pi\alpha\lambda}} \left(\frac{e^\alpha}{e\alpha^\alpha} \right)^\lambda.$$

Note that this reduces to

$$(1.7.6) \quad P(X = x) \sim \frac{1}{\sqrt{2\pi\lambda}}$$

when $\alpha = 1$.

(ii) Show that $\gamma = e^\alpha/e\alpha^\alpha$ satisfies $0 < \gamma \leq 1$ for all $\alpha > 0$, and that $\gamma = 1$ if and only if $\alpha = 1$.

(iii) Make a table showing the accuracy of the approximations (1.7.5) and (1.7.6) for selected values of α and λ .

2.11 Let X have the negative binomial distribution $Nb(p, m)$ shown in Table 1.6.2. For $p = p_m$ satisfying

$$m(1 - p_m) \rightarrow \lambda > 0 \text{ as } m \rightarrow \infty,$$

show that the negative binomial distribution tends to the Poisson distribution $P(\lambda)$.

2.12 Obtain approximations analogous to those of Problem 2.10 for the probability $P\left(\frac{X}{m} = \alpha \frac{q}{p}\right)$ when $\alpha = 1$ and $\alpha \neq 1$.

2.13 Provide an approximation analogous to (1.2.8) for the case that p is close to 1 rather than to 0.

[**Hint:** If p is close to 1, $q = 1 - p$ is close to 0.]

2.14 The sequences $p_n = 1/100$, $x_n = n/100$, $n = 100, 200, 300, \dots$ and $x_n = 1$, $p_n = 1/n$, $n = 100, 200, 300, \dots$ provide two embeddings for the triple $n = 100$, $x = 1$, $p = 1/100$.

(i) Show that the following provides another such embedding.

$$p_n = \frac{1}{\frac{n}{2} + 50}, \quad x_n = \frac{n^2}{10,000}, \quad n = 100, 200, 300, \dots$$

(ii) Give examples of two additional such embeddings.

Section 3

- 3.1 (i) If $0 \leq a_i \leq b_i$ for all i and if $\sum b_i$ converges, then so does $\sum a_i$ and $\sum a_i \leq \sum b_i$.
 (ii) If $\sum a_i$ and $\sum b_i$ converge, then so do $\sum (a_i + b_i)$ and $\sum (a_i - b_i)$ and $\sum (a_i \pm b_i) = \sum a_i \pm \sum b_i$.
- 3.2 If $\sum |a_i|$ converges, so does $\sum a_i$

[**Hint:** Let

$$b_i = \begin{cases} a_i & \text{if } a_i \geq 0 \\ 0 & \text{otherwise} \end{cases} \quad \text{and} \quad c_i = \begin{cases} -a_i & \text{if } a_i \leq 0 \\ 0 & \text{otherwise} \end{cases}.$$

Then $\sum b_i$ and $\sum c_i$ converge by part (i) of Problem 3.1 and hence $\sum a_i = \sum (b_i - c_i)$ converges by part (ii).]

- 3.3 If $\sum a_i$ converges, then $a_i \rightarrow 0$ as $i \rightarrow \infty$.

[**Hint:** Consider first the case that all $a_i \geq 0$. Then if a_i does not tend to 0, there exists $\epsilon > 0$ such that infinitely many of the a_i 's are $> \epsilon$, and hence $\sum a_i = \infty$. The case of arbitrary a_i can be reduced to that of non-negative a_i by the device used in the preceding problem.]

- 3.4 If $a_i = i^i$, the sum $\sum a_i x^i$ converges for no $x \neq 0$, i.e., the radius of convergence of the series is 0.

[**Hint:** $a_i x^i = (ix)^i$ and the result follows from the preceding problem.]

The next set of problems utilize the following result.

Theorem 1.7.1 *If*

$$(1.7.7) \quad f(x) = \sum_{i=0}^{\infty} a_i x^i \quad \text{converges for all } |x| < d,$$

then for all $|x| < d$, the derivative of f is given by

$$(1.7.8) \quad f'(x) = \sum_{i=1}^{\infty} i a_i x^{i-1}$$

and the right side of (1.7.8) converges for all $|x| < d$.

- 3.5 Use Theorem 1.7.1 to show that if (1.7.7) holds, then the second derivative of f is

$$(1.7.9) \quad f''(x) = \sum_{i=2}^{\infty} i(i-1) a_i x^{i-2} \quad \text{for all } |x| < d.$$

- 3.6 (i) If X has the Poisson distribution $P(\lambda)$, determine $E(X)$ and $E[X(X-1)]$.
 (ii) Use (i) to find $\text{Var}(X)$.
 (iii) Determine the 3rd central moment $\mu_3 = E(X-\lambda)^3$.
 [Hint for (i): Use the fact that $E(X) = \sum_{i=1}^{\infty} \frac{i\lambda^i}{i!}$, the exponential series (1.3.9), and (1.7.8).]
- 3.7 (i) Obtain $E(X)$, $\text{Var}(X)$, and $E[X - E(X)]^3$ when X has the negative binomial distribution (1.3.12) instead of the Poisson distribution.
 (ii) Specialize the results of (i) to the geometric distribution (1.3.13).
- 3.8 Solve Problem 3.7(i) when X has the logarithmic distribution (1.3.15).
- 3.9 Show that

$$1 + x + \cdots + x^{n-1} = \frac{x^n - 1}{x - 1}.$$

[Hint: Multiply both sides by $x - 1$.]

Section 4

- 4.1 Prove the bias-variance decomposition (1.4.6).
 [Hint: On the left side add and subtract $E[\delta(X)]$ inside the square bracket.]
- 4.2 In Table 1.4.4, add the rows n^{10} and n^{20} .
- 4.3 Prove parts (i)–(v) of Lemma 1.4.1.
- 4.4 (i) If $a_n = o(c_n)$ and $b_n = o(c_n)$, then $a_n + b_n = o(c_n)$.
 (ii) If a_n, b_n, c_n, d_n are all > 0 , then $a_n = o(c_n)$ and $b_n = o(d_n)$ implies $a_n + b_n = o(c_n + d_n)$.
 (iii) The conclusion of (ii) is no longer valid if the a 's, b 's, c 's, and d 's are not all positive.
 (iv) If $a_n = o(b_n)$, then $a_n + b_n \sim b_n$.
- 4.5 Consider the three statements:
- (1) $b_n \sim a_n$,
 - (2) $b_n = a_n + o(a_n)$,
 - (3) $b_n = a_n + o(1)$.

For each pair (i, j) , determine whether (i) implies (j) , (j) implies (i) , or whether neither implication holds.

4.6 Replace each of the following quantities by a simpler one which consists of only one of the terms and is asymptotically equivalent:

- (i) $\log n + \frac{1}{2}n$;
- (ii) $\log n + \log(\log n)$;
- (iii) $n^2 + e^n$.

4.7 (i) Prove (1.4.17).

(ii) Prove that $n^k = o\left(e^{an^b}\right)$ for any $a, b > 0$.

4.8 Show that

$$\left[1 + \frac{c}{n} + o\left(\frac{1}{n}\right)\right]^n \rightarrow e^c \text{ as } n \rightarrow \infty.$$

[**Hint:** Given any $\epsilon > 0$, there exists n_0 such that $1 + \frac{c - \epsilon}{n} < 1 + \frac{c}{n} + o\left(\frac{1}{n}\right) < 1 + \frac{c + \epsilon}{n}$ for all $n > n_0$.]

4.9 A sum of S dollars is deposited in a checking account which pays $p\%$ interest per annum. If the interest is compounded monthly, determine the value of the account after n months.

4.10 Show that the following sequences tend to infinity as $n \rightarrow \infty$:

- (i) $\log n$;
- (ii) $\log \log n$.

4.11 If $a_n/b_n \rightarrow l$ ($0 < l < \infty$), then $a_n \asymp b_n$.

4.12 State and prove the analogs of the five parts of Lemma 1.4.1 when $a_n = o(b_n)$ is replaced by $a_n \asymp b_n$.

4.13 For each of the following functions f , determine whether $f(cn) \asymp [f(dn)]$ if c and d are any positive constants:

- (i) $f(n) = 2n$,
- (ii) $f(n) = n^2$,
- (iii) $f(n) = e^n$,
- (iv) $f(n) = \log n$.

4.14 If $b_n \rightarrow 0$, and a_n and b_n are both of the same order as c_n , what can you say about the relation of a_n and b_n ?

4.15 Examine the validity of the four parts of Problem 4.4 with o replaced by $\asymp$.

- 4.16 Show that $a_n \asymp b_n$ if and only if both $a_n = O(b_n)$ and $b_n = O(a_n)$.
- 4.17 State and prove the analogs of the five parts of Lemma 1.4.1 with o replaced by O .
- 4.18

$$\text{Let } \begin{array}{l} a_{1n} \text{ and } A_{1n} \text{ be } o(1/n) \\ a_{2n} \text{ and } A_{2n} \text{ be } o(1) \\ a_{3n} \text{ and } A_{3n} \text{ be } o(n) \end{array} \quad \left| \quad \begin{array}{l} a_{4n} \text{ and } A_{4n} \text{ be } O(1/n) \\ a_{5n} \text{ and } A_{5n} \text{ be } O(1) \\ a_{6n} \text{ and } A_{6n} \text{ be } O(n) \end{array} \right.$$

For each of the sums $a_{in} + A_{jn}$, determine the best possible order statement.

[**Hint:** For example, $a_{1n} + A_{1n} = o(1/n)$.]

- 4.19 In the preceding problem, assume that all the a_{in} and A_{jn} are positive and determine the best possible order statement for each product $a_{in}A_{jn}$.
- 4.20 In Example 1.4.6, verify the statements (i)–(iv).
- 4.21 For the two statements

$$a_n = o(n^\alpha), \quad a_n = O(n^\beta),$$

determine whether either implies the other, and hence which is more informative when

- (i) $0 < \alpha < \beta$,
 (ii) $\beta < \alpha < 0$.

- 4.22 Determine the order relation between

$$a_n = \sqrt{\log n} \text{ and } b_n = \log(\sqrt{n})$$

Section 5

- 5.1 Use Definition 1.5.1 to show that $\sqrt{x} \rightarrow \infty$ as $x \rightarrow \infty$.
- 5.2 Use Definition 1.5.2 to prove the second statement of (1.5.5).
- 5.3 There exist values $x > 0$ arbitrarily close to 0 for which $\sin(1/x)$
- (i) equals +1,
 (ii) equals -1.

[**Hint:** $\sin y = 1$ for all $y = \frac{\pi}{2} + k \cdot 2\pi$, $k = 1, 2, \dots$]

- 5.4 Determine the behavior of $\cos(1/x)$ as $x \rightarrow 0+$.

5.5 Let $f(x) = [x]$ denote the greatest integer $\leq x$ (sometimes called the integral part of x). Determine at which values of x , f is continuous, continuous from the right, and continuous from the left.

5.6 A function f is said to be differentiable at x if

$$(1.7.10) \quad \lim_{\Delta \rightarrow 0} \frac{f(x + \Delta) - f(x)}{\Delta}$$

exists, and this limit is then called the derivative of f at x and denoted by $f'(x)$. If f is differentiable at x , then

- (i) it is continuous at x ,
- (ii) an alternative expression for $f'(x)$ is

$$(1.7.11) \quad f'(x) = \lim_{\Delta \rightarrow 0} \frac{f(x + \Delta) - f(x - \Delta)}{2\Delta}.$$

5.7 The following example shows that the existence of the limit (1.7.11) does not necessarily imply that of (1.7.10). Let $f(x) = \min(x, 1 - x)$. Determine for each $0 < x < 1$ whether

- (i) f is continuous,
- (ii) the limit (1.7.10) exists,
- (iii) the limit (1.7.11) exists.

5.8 Define

- (i) $f(x) \sim g(x)$ as $x \rightarrow x_0$,
- (ii) $f(x) = O[g(x)]$ as $x \rightarrow x_0$.

5.9 Show that the results of Examples 1.4.3–1.4.5 remain valid if n is replaced by a continuous variable x .

5.10 (i) If X has the binomial distribution (1.2.1), graph the function $f(x) = P(X \leq x)$ for the case $n = 2$, $p = 1/2$. At its discontinuities, is F continuous on the right, the left, or neither?

- (ii) Solve part (i) if $F(x) = P(X < x)$.

5.11 If f is non-decreasing on $a < x < x_0$, then as x tends to x_0 from the left, $f(x)$ tends either to a finite limit or to ∞ .

[**Hint:** (i) If $f(x_0) < \infty$, then f is bounded above for $x < x_0$ by $f(x_0)$. If m is the least upper bound of $f(x)$ for $x < x_0$, then $f(x) \rightarrow m$ as $x \rightarrow x_0$ from the left.

- (ii) If $f(x)$ is not bounded above for $x < x_0$, then $f(x) \rightarrow \infty$ as $x \rightarrow x_0 -$.]

5.12 Give an example of each of the following three possibilities for a set S with least upper bound m :

- (i) m is a member of S but not a limit point of S ;
- (ii) m is a limit point of S but not a member of S ;
- (iii) m is both a member and a limit point of S .

5.13 Determine the inf and the sup of each of the following sets and determine whether it is in the set.

- (i) $S = \left\{ 1 + \frac{1}{n}, \quad n = 1, 2, \dots \right\}$;
- (ii) $S = \left\{ 1 + (-1)^n \frac{1}{n}, \quad n = 1, 2, \dots \right\}$;
- (iii) $S = \left\{ \frac{1}{\sqrt[n]{2} - 1}, \quad n = 1, 2, \dots \right\}$.

5.14 If $\{a_n\}$ and $\{b_n\}$ are two sequences of numbers with $a_n \leq b_n$ for all n and $b_n \rightarrow b$, then $\sup \{a_n\} \leq b$.

5.15 Let

$$f(x) = \begin{cases} 1 & \text{if } x = 1/n, \quad n = 1, 2, \dots \\ 0 & \text{for all other values of } x. \end{cases}$$

Determine for which values of x the function f is continuous.

Section 6

6.1 Let $F^*(x) = \frac{1}{2} [P(X \leq x) + P(X < x)]$. At a point of discontinuity of F , determine whether F^* is continuous on the left, on the right, or neither.

6.2 For any cdf F , the quantile function F^{-1} defined by (1.6.10) is

- (i) nondecreasing;
- (ii) continuous on the left.

6.3 For any cdf F , we have

- (i) $F^{-1}[F(x)] \leq x$ for all $-\infty < x < \infty$;
- (ii) $F[F^{-1}(y)] \geq y$ for all $0 < y < 1$.

6.4 Use the preceding problem to show that for any F ,

$$(1.7.12) \quad F(x) \geq y \quad \text{if and only if} \quad x \geq F^{-1}(y).$$

6.5 If F is continuous at x and $0 < y = F(x) < 1$, then

- (i) $F[F^{-1}(y)] = y$ but
- (ii) $F^{-1}[F(x)]$ may be $< x$.

[**Hint (ii)**: Let x be the right-hand end point of a flat stretch of F .]

6.6 (i) The cdf of the logistic distribution $L(0, 1)$ given in Table 1.6.1 is

$$(1.7.13) \quad F(x) = \frac{1}{1 + e^{-x}}.$$

- (ii) Give an explicit expression for F^{-1} .

[**Hint (i)**: Differentiate (1.7.13).]

6.7 (i) The cdf of the exponential distribution $E(0, 1)$ given in Table 1.6.1 is

$$(1.7.14) \quad F(x) = 1 - e^{-x}.$$

- (ii) Give an explicit expression for F^{-1} .

6.8 If X is distributed as $U(0, 1)$, then $-\log X$ is distributed as $E(0, 1)$.

6.9 If X has the continuous cdf F , then $F(X)$ is distributed as $U(0, 1)$.

[**Hint**: Use (1.7.12).]

6.10 If the graph of F is rotated about the line $y = x$, the resulting inverse F^{-1} is continuous on the left.

[**Hint**: Consider how south-north and west-east stretches of F (corresponding to discontinuities and flat stretches, respectively) are transformed under the rotation.]

Bibliographic Notes

Among the many books providing a rigorous treatment of basic concepts such as limit, continuity, orders of magnitude, and the convergence of infinite series, two outstanding classical texts (both recently reissued) are Hardy (1908, 1952, 1992) and Courant (1927, 1937, 1982). A very accessible presentation in the more general setting of metric spaces is Rudin (1976). Knopp (1922, 1951, 1990) provides an extensive account of infinite series, and the first chapter of de Bruijn (1958) contains a general introduction to asymptotics with a detailed discussion of the o, O notation. Details on the distributions of Table 1.6.2 and others can be found in Johnson, Kotz, and Kemp (1992) and on those in Table 1.6.1 in Johnson, Kotz, and Balakrishnan (1994, 1995).